

Introduction to Machine Learning

Week 1

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1. (2 Marks) Which of the following are supervised learning problems (Multiple Correct)?

- (a) Clustering Spotify users based on their listening history
- (b) Weather forecast using data collected by a satellite
- (c) Predicting tuberculosis using patient's chest X-Ray
- (d) Training a humanoid to walk using a reward system

Sol. b and c

2. (2 Marks) Which of the following are regression tasks (Multiple Correct)?

- (a) Predicting the outcome of an election
- (b) Predicting the weight of a giraffe based on its weight
- (c) Predicting the emotion conveyed by a sentence
- (d) Identifying abnormal data points

Sol. b

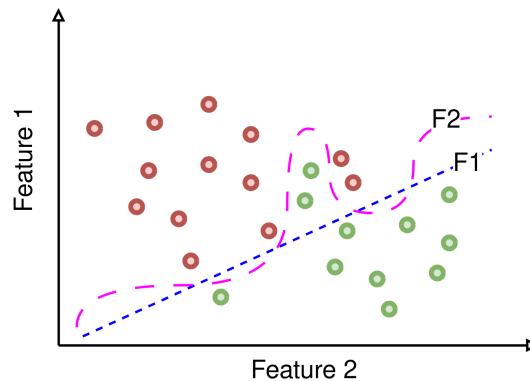
3. (2 Marks) Which of the following are classification tasks (Multiple Correct)?

- (a) Predicting the outcome of an election
- (b) Predicting the weight of a giraffe based on its weight
- (c) Predicting the emotion conveyed by a sentence
- (d) Identifying abnormal data points

Sol. a, and c

(Common Data for Questions 4 and 5)

Here is a 2-dimensional plot showing two functions that classify data points into two classes. The red points belong to one class, and the green points belong to another. The dotted blue line (F1) and dashed pink line (F2) represent the two trained functions.



4. (1 Mark) Which of the two functions overfit the training data?

- (a) Both functions F1 & F2
- (b) Function F1
- (c) Function F2
- (d) None of them

Sol. c

5. (1 Mark) Which of the following 2 functions will yield higher training error?

- (a) Function F1
- (b) Function F2
- (c) Both functions F1 & F2 will have the same training error
- (d) Can not be determined

Sol. a

6. (1 Mark) What does the term ‘policy’ refer to in reinforcement learning?

- (a) A set of rules governing the environment
- (b) The reward function
- (c) The initial state of the environment
- (d) The strategy the agent follows to choose actions

Sol. d

7. (1 Mark) Given the following dataset, for $k = 3$, use KNN regression to find the prediction for a new data-point (2,3) (*Use Euclidean distance measure for finding closest points*)

X1	X2	Y
2	5	3.4
5	5	5
3	3	3
6	3	4.5
2	2	2
4	1	2.8

- (a) 2.0
- (b) 2.6
- (c) 2.8
- (d) 3.2

Sol. c: The closest k points are (2,5), (3,3) and (2,2). Their corresponding labels averaged is $(3.4 + 3 + 2) / 3 = 2.8$

8. (1 Mark) For any given dataset, comment on the bias of K-nearest classifiers upon increasing the value of K .

- (a) The bias of the classifier decreases
- (b) The bias of the classifier does not change
- (c) The bias of the classifier increases
- (d) Can not be determined

Sol. c: Refer to lecture

9. (1 Mark) Bias and variance are given by:

- (a) $\mathbb{E}[\hat{f}(x)] - f(x), \mathbb{E}[(\mathbb{E}[\hat{f}(x)] - \hat{f}(x))^2]$
- (b) $\mathbb{E}[\hat{f}(x)] - f(x), \mathbb{E}[(\mathbb{E}[\hat{f}(x)] - \hat{f}(x))]^2$
- (c) $(\mathbb{E}[\hat{f}(x)] - f(x))^2, \mathbb{E}[(\mathbb{E}[\hat{f}(x)] - \hat{f}(x))^2]$
- (d) $(\mathbb{E}[\hat{f}(x)] - f(x))^2, \mathbb{E}[(\mathbb{E}[\hat{f}(x)] - \hat{f}(x))]^2$

Sol. a

10. (1 Mark) Which of the following statements are FALSE regarding bias and variance?

- (a) Models which overfit have a high bias
- (b) Models which overfit have a low bias
- (c) Models which underfit have a high variance
- (d) Models which underfit have a low variance

Sol. a and c